

Initial semantics of incompletely specified actions: forced quotients, kernel saturation, and completion

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Abstract

An incompletely specified action need not admit a completion on its original carrier, and its unspecified values need not remain independent. We study finite two-sorted ground presentations consisting of complete native operation tables, carrier-compatibility equations, and arbitrary supplied action cells. For this pure-row fragment, we characterize the carrier congruence forced by initial semantics as the least quotient on which every row pushout preserves the distinguished carrier. Row protection is equivalent to homomorphic coherence of the supplied data and an exact kernel-extension condition. On the resulting quotient, if a row induces $h : S \rightarrow Q$ and $\kappa = \text{Cg}_Q(\ker h)$, its forced domain is precisely the saturation S^κ . We identify the full equality relation on the named carrier and named action cells and give its exact class counts at proof horizons one and two. Carrier-valued completion has a different criterion: the carrier embedding into the row pushout must split. Finite examples show that repairs need not have a least element, need not be upward closed, and can force strictly more carrier equality than unrestricted models. Further results describe the effect of composition, group laws, and one-sorted coupling. The proofs use unrestricted ground-diagram semantics; no identities of a chosen variety are imposed on unnamed elements.

Keywords. Initial algebra; ground equations; incomplete action; congruence extension; pushout; kernel saturation; quotient repair.

1 Introduction

Suppose that a finite carrier algebra A and a finite operator algebra B are known completely, but only some values of an operation $\alpha : B \times A \rightarrow A$ have been prescribed. Requiring each named operator to preserve the native operations of A connects the missing cells to one another. It can also force originally distinct carrier elements to become equal. A semantics that immediately fills every blank with a carrier element can therefore obscure what the original equations actually determine.

This paper separates three questions. Which equalities among the original carrier names hold in every model? Which missing action values already equal carrier names in every model? On which carrier quotients can all missing values be chosen inside the quotient? These are, respectively, questions about *protection*, *determinacy*, and *completion*. They have different answers, even for a single row on a four-element algebra.

We use initial-algebra semantics for a finite ground presentation. The native tables specify named copies of A and B ; compatibility is required only at named inputs. An unspecified cell denotes a total operation value in every model, but that value need not belong to the named carrier image. We call such an initial value *external*. This use of external values is essential to distinguish necessary identifications from choices introduced by a completion.

The main results are as follows.

1. **Exact forced quotient.** Each row is a homomorphic image of the carrier, glued to it along the supplied data. A row pushout preserves the carrier exactly when the data induce a homomorphism $h : S \rightarrow Q$ whose kernel extends to Q without acquiring new pairs inside S (Theorem 3.4). The least quotient with this property for every row is the exact carrier quotient of the initial algebra (Theorem 5.2). A monotone feedback operator computes it in at most $|A| - 1$ strict quotient steps, and this bound is sharp.
2. **Exact forced domain.** For a protected row, the ambient congruence $\kappa = \text{Cg}_Q(\ker h)$ is exactly the row kernel. A point has a forced carrier value precisely when its κ -class meets S (Theorem 4.1). Theorem 5.3 transfers this description to all rows in the full initial algebra. The resulting description also gives exact class counts on the named interface at proof horizons one and two (Theorem 6.6).
3. **Completion and repair.** Completing a row on Q is equivalent to splitting its carrier embedding into the pushout (Theorem 7.1). The repair congruences are exactly the carrier kernels of models whose named action values stay inside the named carrier image (Theorem 7.7). Examples separate this model restriction from unrestricted equational consequence.

The categorical and logical tools are classical: initial algebras, amalgamation, congruence generation, and ground congruence closure. The contribution is their exact combination for arbitrary incomplete action rows, including inconsistent supplied values. In particular, the input is not assumed to be a partial action or even a partial function. The homomorphic partial maps arise after the required quotient has been found.

This article is the semantic foundation of a three-paper programme. The companion manuscript on equality-visibility depth [20] studies the dependence of finite-ground equality on a chosen presentation. The planned third paper concerns the computational core and checkable certificates. Here we retain only the finite-ground filtration needed to connect the structural results to those later questions. All results used in this paper are proved here or attributed to the preceding literature; the argument does not depend on either companion being published.

Sections 2–5 develop the pure-row theory. Section 6 gives the examples and named-interface filtration. Section 7 treats completion and repair. Sections 8 and 9 delimit the effect of additional action laws and one-sorted coupling. Section 10 explains finite verification, and Section 11 situates the results in the literature.

2 Typed ground action specifications

2.1 Native diagrams and the action symbol

All algebras are nonempty and all native operation signatures are finite and finitary. Elements are named by designated constants; native operation symbols are assumed to have positive arity. Let

$$A = (|A|, \Sigma_A), \quad B = (|B|, \Sigma_B)$$

be finite algebras. The two-sorted signature has sorts $\mathbf{A}$ and $\mathbf{B}$, native operations of Σ_A on $\mathbf{A}$, native operations of Σ_B on $\mathbf{B}$, designated constants $c_a : \mathbf{A}$ for $a \in A$ and $c_b : \mathbf{B}$ for $b \in B$, and

$$\alpha : \mathbf{B} \times \mathbf{A} \rightarrow \mathbf{A}.$$

We usually suppress the constant symbols and write a, b for their names.

The complete native ground diagrams are

$$D_A = \{f(a_1, \dots, a_k) = f^A(a_1, \dots, a_k)\},$$

with one row for every $f \in \Sigma_A$ and named tuple, and similarly D_B . The base action theory contains the ground carrier-compatibility instances

$$\alpha(b, f(a_1, \dots, a_k)) = f(\alpha(b, a_1), \dots, \alpha(b, a_k)) \quad (2.1)$$

for named $b \in B$ and named $a_i \in A$.

An incomplete action table is an arbitrary finite set of row occurrences

$$\Gamma \subseteq \{\alpha(b, a) = c : b \in B, a, c \in A\}. \quad (2.2)$$

Occurrences, rather than a partial function, are allowed: two supplied equations may have the same input and different outputs. A complete table is the special case in which one output is supplied for every (b, a) .

Sections 3–7 study the *pure row fragment*

$$E = D_A \cup D_B \cup \text{Comp}_A \cup \Gamma,$$

where Comp_A denotes the family (2.1). Ground action laws such as unit, composition, and inverse are added later.

These are equations between closed terms, not universally quantified identities with unnamed inputs. Models range over all algebras of the specified two-sorted signature. In particular, an identity that happens to hold in A is not automatically required on the whole carrier sort of a model. The operator sort of the initial algebra is exactly its named copy of B : every closed operator term reduces by D_B , and the model constructions below separate its names.

2.2 Initial algebra, carrier quotient, and external values

Let $\mathbb{I}_E$ be the initial two-sorted algebra of the equational theory E . The map from carrier names into the A-sort of $\mathbb{I}_E$ need not be injective. Define the induced carrier congruence

$$\theta_E^A = \{(a, a') \in A^2 : E \vdash a = a'\}. \quad (2.3)$$

Since provable equality is a congruence, $\theta_E^A \in \text{Con}(A)$. Put

$$Q_E = A/\theta_E^A.$$

The named carrier embeds canonically as Q_E in the initial A-sort.

Definition 2.1. For a named $b \in B$ and $a \in A$, the class of $\alpha(b, a)$ is *carrier-valued* if it contains a carrier name. Otherwise it is *external*. The named ground action is *closed* if every $\alpha(b, a)$ is carrier-valued.

The three central diagnostics are now formal.

Definition 2.2 (Protection, determinacy, realizability). The presentation is *carrier-protecting* when $\theta_E^A = \Delta_A$. For a fixed row b , its *forced domain* in Q_E is

$$D_b = \{[a] \in Q_E : \exists c \in A \text{ with } E \vdash \alpha(b, a) = c\}.$$

A *carrier-valued completion* on a quotient A/θ is a family of endomorphisms

$$\tau_b \in \text{End}(A/\theta) \quad (b \in B)$$

satisfying all projected supplied rows. Additional action laws, when present, impose the corresponding equations on the family (τ_b) .

The endomorphism requirement is exactly carrier compatibility: a total row map $\tau_b : Q \rightarrow Q$ satisfies all instances of (2.1) precisely when it preserves every native carrier operation.

Definition 2.3 (Repair set). For a pure-row presentation, define

$$\mathcal{R}(E) = \{\theta \in \text{Con}(A) : \text{the projected rows admit a carrier-valued completion on } A/\theta\}. \quad (2.4)$$

Elements of $\mathcal{R}(E)$ are called *repairs*.

The total congruence is always a repair, so $\mathcal{R}(E)$ is nonempty.

Symbol	Meaning throughout the pure-row sections
θ_E^A, Q_E	Forced carrier congruence and A/θ_E^A .
S_b, h_b	Subalgebra generated by supplied inputs, and its induced row homomorphism on a protected quotient.
K_b, κ_b	$\ker h_b$ and its generated congruence $\text{Cg}_Q(K_b)$ on Q .
D_b	Domain of the carrier-valued part of the initial row.
$\theta^{[r]}$	Carrier congruence after r feedback rounds.
$D, \equiv_D$	Term-depth horizon and equality visible at that horizon.
$\mathcal{I}$	Named carrier and named action interface.
$\mathcal{R}(E)$	Quotients admitting a carrier-valued completion.
κ_E^{NJ}	Intersection of all repair congruences.

3 Rows as homomorphic copies and partial gluings

3.1 Every named row is a homomorphic carrier image

The basic structural observation is immediate but decisive.

Proposition 3.1 (Row-homomorphism principle). *For each named $b \in B$, the map*

$$j_b : A \rightarrow \mathbb{I}_E^A, \quad j_b(a) = [\alpha(b, a)]$$

is a homomorphism of the native carrier signature. It factors through

$$\bar{j}_b : Q_E \rightarrow \mathbb{I}_E^A.$$

Proof. For a native k -ary f and named $a_1, \dots, a_k$, compatibility (2.1) and the row of D_A give

$$j_b(f^A(\bar{a})) = [f(\alpha(b, a_1), \dots, \alpha(b, a_k))] = f^{\mathbb{I}_E}(j_b(a_1), \dots, j_b(a_k)).$$

If $a \equiv_{\theta_E^A} a'$, congruence under $\alpha(b, -)$ gives $j_b(a) = j_b(a')$, so j_b factors through Q_E . $\square$

Thus missing cells do not generate unrelated points. For each operator b they lie in a homomorphic image of the whole carrier quotient. A supplied row $\alpha(b, a) = c$ identifies the row point $\bar{j}_b([a])$ with the distinguished carrier point $[c]$. This is the local gluing picture that will be made exact by pushouts.

3.2 Strong amalgamation in the unrestricted carrier signature

Let Σ_A^+ denote the positive-arity native carrier signature, with the designated names omitted. We use the category

$$\mathbf{Alg}_{\Sigma_A^+}$$

of all algebras of this fixed signature, not the variety generated by A . This distinction matters: the ground diagram specifies the operation table of the named copy but does not impose every identity true in A on arbitrary elements of a model.

We use the following standard free-amalgamation fact.

Lemma 3.2 (Strong amalgamation). *The category $\mathbf{Alg}_\Sigma$ of all algebras of a fixed finitary positive-arity signature has the strong amalgamation property. If embeddings $S \hookrightarrow C$ and $S \hookrightarrow D$ are given, their pushout can be chosen so that both C and D embed and their images intersect exactly in the common image of S .*

Proof. Identify the two copies of S and put $U = C \sqcup_S D$. Construct normal forms starting with the elements of U . For a native operation symbol f and already constructed normal forms $x_1, \dots, x_k$, evaluate the tuple inside C if all x_i belong to C , or inside D if all belong to D . When both rules apply, all arguments lie in S , so the two answers agree. When neither rule applies, retain $f(x_1, \dots, x_k)$ as a new formal node. Finite recursion gives a set of normal forms closed under these operations.

The inclusions of C and D are homomorphisms, remain injective, and have intersection exactly S , since no two distinct base elements were identified. Any two homomorphisms out of C and D agreeing on S extend uniquely by recursive evaluation of formal nodes. This is the pushout universal property. The construction is the standard free amalgam for an unrestricted signature; see also [9, 10]. $\square$

Remark 3.3. Lemma 3.2 would be false in this form for an arbitrary nontrivial variety. In a variety, additional identities can create dominion or amalgamation phenomena. The clean kernel theory below is tied to the ground-diagram semantics of the present paper.

3.3 The raw row pushout

Fix a congruence $\theta \in \text{Con}(A)$ and put $Q = A/\theta$. Fix a named $b \in B$, and let R_b be the finite set of supplied row occurrences for b . If $R_b \neq \emptyset$, let

$$F_b = F_{\Sigma_A^+}(\{x_r : r \in R_b\})$$

be the absolutely free carrier algebra on one generator for each occurrence. Let $Q_b \cong Q$ be a formal row copy and $Q_* \cong Q$ the distinguished carrier copy. If occurrence r is

$$\alpha(b, a_r) = c_r,$$

define

$$u_b : F_b \rightarrow Q_b, \quad u_b(x_r) = [a_r],$$

and

$$v_b : F_b \rightarrow Q_*, \quad v_b(x_r) = [c_r].$$

Form the pushout

$$\begin{array}{ccc} F_b & \xrightarrow{u_b} & Q_b \\ v_b \downarrow & & \downarrow p_b \\ Q_* & \xrightarrow{m_b} & P_b. \end{array} \tag{3.1}$$

If $R_b = \emptyset$, use the coproduct of the two disjoint copies of Q . The same normal-form construction applies without a common subalgebra; the positive-arity convention is relevant here. The row is automatically protected at this stage.

Let

$$S_b = \text{im } u_b = \langle [a_r] : r \in R_b \rangle_{Q}.$$

The next theorem gives the exact local obstruction.

Theorem 3.4 (Raw row protection criterion). *Assume $R_b \neq \emptyset$. The carrier leg $m_b : Q_* \rightarrow P_b$ in (3.1) is injective if and only if both of the following hold.*

(a) Homomorphic coherence:

$$\ker u_b \subseteq \ker v_b.$$

Equivalently, the supplied values induce a unique homomorphism

$$h_b : S_b \rightarrow Q$$

with $h_b u_b = v_b$ after corestricting u_b to S_b .

(b) Kernel-extension stability: *with*

$$K_b = \ker h_b, \quad \kappa_b = \text{Cg}_Q(K_b),$$

one has

$$\kappa_b \cap S_b^2 = K_b. \tag{3.2}$$

Proof. Suppose first that m_b is injective. If $u_b(x) = u_b(y)$, then in the pushout

$$m_b v_b(x) = p_b u_b(x) = p_b u_b(y) = m_b v_b(y).$$

Injectivity gives $v_b(x) = v_b(y)$, so $\ker u_b \subseteq \ker v_b$. Hence v_b factors uniquely through S_b , defining h_b .

Write $u_b = i_b e_b$ with $e_b : F_b \rightarrow S_b$ and $i_b : S_b \hookrightarrow Q_b$. Since e_b is surjective, a cocone over (u_b, v_b) is the same thing as a cocone over (i_b, h_b) ; thus (3.1) is also the pushout of

$$S_b \hookrightarrow Q_b, \quad h_b : S_b \rightarrow Q_*.$$

Factor h_b through its kernel:

$$S_b \twoheadrightarrow S_b/K_b \hookrightarrow Q_*.$$

Pushing the first map through $S_b \hookrightarrow Q_b$ quotients the row copy by

$$\kappa_b = \text{Cg}_Q(K_b).$$

The induced map $S_b/K_b \rightarrow Q_b/\kappa_b$ is injective exactly when (3.2) holds. If it is injective, the remaining pushout is a strong amalgam of two embeddings, so Lemma 3.2 preserves the central carrier. Conversely, if (3.2) fails, two distinct K_b -classes from S_b already merge in Q_b/κ_b ; the pushout relation then forces their h_b -images to merge in Q_* , contradicting injectivity of m_b . $\square$

The first obstruction includes ordinary value conflicts and all compatibility consequences on the subalgebra generated by the supplied inputs. The second is a particular congruence-extension obstruction: even a genuine partial homomorphism can force additional equality if its kernel does not extend cleanly to the whole carrier.

4 Kernel saturation and exact determinacy

Assume now that Q is a carrier quotient for which a row is protected, and suppress the subscript b . Thus $S \leq Q$, $h : S \rightarrow Q$ is a homomorphism,

$$K = \ker h, \quad \kappa = \text{Cg}_Q(K),$$

and $\kappa \cap S^2 = K$. Let

$$\begin{array}{ccc} S & \hookrightarrow & Q_r \\ h \downarrow & & \downarrow p_r \\ Q_* & \xrightarrow{m} & P \end{array}$$

be the stabilized row pushout.

Theorem 4.1 (Kernel-saturation theorem). *In the protected row pushout above:*

(i) $\ker p_r = \kappa$;

(ii)

$$p_r(Q) \cap m(Q) = m(h(S));$$

(iii) for $x \in Q$,

$$p_r(x) \in m(Q) \iff [x]_\kappa \cap S \neq \emptyset; \quad (4.1)$$

(iv) if $x \equiv_\kappa s$ with $s \in S$, then

$$p_r(x) = m(h(s)).$$

Consequently the exact carrier-forced domain of the row is

$$D = S^\kappa := \{x \in Q : [x]_\kappa \cap S \neq \emptyset\}. \quad (4.2)$$

Proof. Push out $S \hookrightarrow Q_r$ first along the quotient $S \twoheadrightarrow S/K$. The result is $Q_r \twoheadrightarrow Q_r/\kappa$, and the induced copy of S/K is embedded because $\kappa \cap S^2 = K$. The second stage is therefore the strong amalgam of the embeddings

$$S/K \hookrightarrow Q_r/\kappa, \quad S/K \hookrightarrow Q_*.$$

Strong amalgamation gives both the row kernel and the exact intersection of the two images. Statement (4.1) is precisely the condition that the κ -class of x belongs to the embedded copy of S/K . $\square$

Corollary 4.2 (No hidden forcing for injective data). *If $h : S \rightarrow Q$ is injective, then $\kappa = \Delta_Q$ and*

$$D = S.$$

Thus an injective supplied partial homomorphism forces no additional row values outside its generated input subalgebra.

Corollary 4.3 (Exact number of external row classes). *In a finite protected row,*

$$|p_r(Q) \setminus m(Q)| = |Q/\kappa| - |S/K|.$$

For an entirely unspecified row the contribution is $|Q|$.

Proof. By Theorem 4.1, $p_r(Q) \cong Q/\kappa$ and its intersection with $m(Q)$ is isomorphic to $h(S) \cong S/K$. $\square$

The formula shows that “missing” and “free” are not synonymous. Nontrivial kernel propagation can make D strictly larger than S , even equal to all of Q .

5 The exact carrier quotient for an arbitrary incomplete table

5.1 Row-stable congruences

For $\theta \in \text{Con}(A)$ let $Q = A/\theta$ and project every supplied row occurrence to Q . For each named b , form the raw row pushout (3.1) over Q .

Definition 5.1 (Row stability). A congruence $\theta \in \text{Con}(A)$ is *row-stable* if, for every named $b \in B$, the distinguished carrier leg of the corresponding raw row pushout over A/θ is injective. Equivalently, every row satisfies the two conditions of Theorem 3.4.

Theorem 5.2 (Least row-stable carrier quotient). *For the pure row fragment,*

$$\boxed{\theta_E^A = \min\{\theta \in \text{Con}(A) : \theta \text{ is row-stable}\}.} \quad (5.1)$$

In particular, the full carrier congruence forced by an arbitrary incomplete action table is determined entirely inside the finite congruence lattice of A .

Proof. Put $\theta = \theta_E^A$ and $Q = Q_E$. The named carrier quotient Q embeds in the initial $\mathbf{A}$ -sort. For every row, Proposition 3.1 and the supplied cells realize the corresponding gluing inside that initial algebra while preserving Q . By the universal property of the raw pushout, its carrier leg must therefore be injective. Thus θ_E^A is row-stable.

Conversely let θ be row-stable and put $Q = A/\theta$. For each b , Theorem 3.4 gives a homomorphism $h_b : S_b \rightarrow Q$ whose generated kernel κ_b satisfies the extension condition. Factor the row copy by κ_b and strongly amalgamate Q/κ_b with the distinguished Q along the common copy of $S_b/\ker h_b$. Do this successively for all finitely many rows, always over the same embedded distinguished copy of Q . Lemma 3.2 preserves that central copy at every step.

Use the resulting amalgam as the carrier sort of a model. Interpret the operator sort exactly as B . For a named b and a central element $[a] \in Q$, define $\alpha(b, [a])$ to be the corresponding point of the b -row image; define all remaining action values arbitrarily. Native operations are those of the amalgam. Carrier compatibility holds on the named central inputs because each row image is a homomorphic quotient of Q , and the supplied cells hold by construction. Hence this is a model of E whose kernel on the named carrier is exactly θ .

Every provable carrier equality must hold in this model, so $\theta_E^A \subseteq \theta$. Since θ_E^A itself is row-stable, it is the least such congruence. $\square$

Theorem 5.2 is the structural center of the paper. It separates two effects that are conflated by direct table completion. The quotient θ_E^A is forced by unrestricted equational semantics; only after this quotient has been taken does each row possess a canonical forced partial homomorphism, whose exact domain is given by Theorem 4.1.

5.2 All rows inside the full initial algebra

The preceding proof also supplies the separation model needed to transfer local pushout geometry to the full presentation. This transfer is important: a local upper bound on forcing would not by itself rule out extra identifications caused by other rows.

Theorem 5.3 (Exact global row geometry). *Let $Q = Q_E$ and let $\iota : Q \hookrightarrow \mathbb{I}_E^A$ be the named carrier embedding. For each nonempty row, form S_b, h_b, K_b, κ_b over Q . Then*

$$\ker \bar{j}_b = \kappa_b, \quad (5.2)$$

$$\bar{j}_b(Q) \cap \iota(Q) = \iota(h_b(S_b)), \quad (5.3)$$

$$D_b = S_b^{\kappa_b}. \quad (5.4)$$

If $x \equiv_{\kappa_b} s$ with $s \in S_b$, its forced value is $h_b(s)$. The external parts of distinct named rows are disjoint in the initial algebra. For an empty row, the row kernel is diagonal, its image is disjoint from the carrier and all other row images, and $D_b = \emptyset$.

Proof. Theorem 5.2 makes every row protected over Q . The supplied equations imply $\bar{j}_b(s) = \iota(h_b(s))$ on S_b . Consequently $K_b \subseteq \ker \bar{j}_b$, and the row-homomorphism principle implies $\kappa_b \subseteq \ker \bar{j}_b$. This proves all equalities claimed by the theorem, including the values on $S_b^{\kappa_b}$.

For exactness, construct a model as in the second half of the proof of Theorem 5.2. At each step attach Q/κ_b by a strong amalgam along the embedded S_b/K_b in the central Q . The new

row meets the entire previously constructed algebra only along that subalgebra of Q . Thus its kernel is exactly κ_b , its carrier intersection is exactly $h_b(S_b)$, and it adds no intersection between external row parts. Attach an empty row by a disjoint coproduct. Interpreting the action on central inputs by these row maps gives a model of all of E . Every pair excluded by the theorem is separated in this one model, so it cannot become equal in the initial algebra. $\square$

5.3 A finite feedback computation

The theorem also gives a canonical closure procedure. For $\theta \in \text{Con}(A)$ let $Q_\theta = A/\theta$, form every raw row pushout over Q_θ , and write

$$\chi_{b,\theta} = \ker(m_{b,\theta}) \in \text{Con}(Q_\theta).$$

Let $\pi_\theta : A \twoheadrightarrow Q_\theta$ and define

$$\Phi(\theta) = \bigvee_{b \in B} \pi_\theta^{-1}(\chi_{b,\theta}). \quad (5.5)$$

For an unspecified row, $\chi_{b,\theta} = \Delta_{Q_\theta}$.

Proposition 5.4 (Pushout feedback operator). *The operator $\Phi : \text{Con}(A) \rightarrow \text{Con}(A)$ is inflationary and monotone. Its fixed points are exactly the row-stable congruences. Consequently*

$$\boxed{\theta_E^A = \text{lfp}(\Phi)}. \quad (5.6)$$

Starting with $\theta^{[0]} = \Delta_A$ and $\theta^{[r+1]} = \Phi(\theta^{[r]})$, at most $|A| - 1$ strict steps occur.

Proof. Each pullback $\pi_\theta^{-1}(\chi_{b,\theta})$ contains θ , giving inflationarity. If $\theta \subseteq \varphi$, the quotient map $A/\theta \twoheadrightarrow A/\varphi$ induces a morphism between the two raw row spans and hence between their pushouts. Equality created in the carrier leg at level θ therefore remains equality after passage to φ , which gives monotonicity after pulling back to A .

The join in (5.5) equals θ precisely when every $\chi_{b,\theta}$ is diagonal, i.e. precisely when every raw row pushout protects the carrier. The fixed-point statement now follows from Theorem 5.2. In a strict step the number of congruence classes decreases by at least one, so a finite A allows at most $|A| - 1$ strict steps. $\square$

Proposition 5.5 (Sharpness of the feedback bound). *For every $n \geq 2$ there is an n -element unary carrier and an incomplete action presentation for which the iteration of Proposition 5.4 has exactly $n - 1$ strict steps.*

Proof. Let $A_n = \{0, 1, \dots, n-1\}$ with one unary operation $f(x) = 0$ for all x . Take distinct operators $b_1, \dots, b_{n-1}$ and supply exactly

$$\alpha(b_i, i-1) = i \quad (1 \leq i < n).$$

At the first stage the b_1 row asks for $h(0) = 1$. A protected homomorphic row would satisfy

$$1 = h(0) = h(f(0)) = f(h(0)) = f(1) = 0,$$

so $0 \equiv 1$ is forced. After quotienting by this equality, the source of the b_2 row is the zero class and the same argument forces $[0] = [2]$. To check that no other collapse occurs in the same round, suppose the current blocks are $\{0, \dots, k\}$ and the remaining singletons. Rows with $i \leq k$ already map the zero class to itself. Row $i = k+1$ forces exactly its target into the zero class. For $i > k+1$, the supplied input generates the two-element subalgebra consisting of the zero class and $[i-1]$; the prescribed extension sends these to the zero class and $[i]$. It is an injective homomorphism and hence protected by Theorem 3.4. Because the update is simultaneous, these later rows contribute no additional equality in that round. Induction therefore gives exactly the block $\{0, \dots, r\}$ after r strict rounds, and total collapse first occurs after $n - 1$ rounds. $\square$

5.4 Two endpoint corollaries

The earlier special cases now become immediate consequences.

Corollary 5.6 (Annihilator row). *Assume designated elements $0_A, 0_B$ are present and that Γ consists only of the complete row*

$$\alpha(0_B, a) = 0_A \quad (a \in A).$$

All other rows are unspecified. Then the exact carrier congruence is

$$\theta_{\text{Ann}} = \text{Cg}_A\{(0, f^A(0, \dots, 0)) : f \in \Sigma_A\}. \quad (5.7)$$

Proof. On a quotient Q , the annihilator row is the constant-zero map $Q \rightarrow Q$. Since the row is complete, stability is exactly the assertion that this map is an endomorphism, i.e. that $[0]$ is closed under every positive-arity carrier operation. The least quotient with that property is (5.7). $\square$

Corollary 5.7 (Complete action table). *Suppose one value*

$$\alpha(b, a) = \tau_b(a)$$

is supplied for every $(b, a) \in B \times A$. Then θ_E^A is the least native carrier congruence θ for which every τ_b descends to an endomorphism of A/θ . Equivalently, it is generated, in the expanded algebra $(A, \Sigma_A, \{\tau_b\}_{b \in B})$, by the compatibility defects

$$(\tau_b(f(\bar{a})), f(\tau_b(a_1), \dots, \tau_b(a_k))).$$

Thus the annihilator and complete-table formulas are not isolated regimes. They are instances of the case $S_b = Q$ of the same arbitrary-partial-row theorem.

6 Kernel effects and the named-interface filtration

6.1 Kernel amplification and kernel obstruction

The next two examples show why homomorphic coherence alone does not describe incomplete rows.

Example 6.1 (One supplied cell forces the whole row). Let $A = \{0, 1, 2\}$ with one binary operation f given by

f	0	1	2
0	0	0	0
1	0	0	0
2	0	1	0

and take one operator b . Supply only

$$\alpha(b, 2) = 0.$$

The generated input subalgebra is $S = \{0, 2\}$. Compatibility gives

$$\alpha(b, 0) = f(\alpha(b, 2), \alpha(b, 2)) = 0.$$

The induced map $h : S \rightarrow A$ is therefore constant zero, so its kernel identifies 0 and 2. But

$$\text{Cg}_A(0, 2) = \nabla_A,$$

because the context $f(-, 1)$ sends 0 and 2 to 0 and 1. Hence Theorem 4.1 gives $D = A$: the single supplied cell forces

$$\alpha(b, 0) = \alpha(b, 1) = \alpha(b, 2) = 0.$$

The carrier nevertheless remains protected, since the constant-zero map is an endomorphism of A .

This is pure determinacy without confusion: kernel propagation enlarges the forced domain while preserving all carrier distinctions.

Example 6.2 (A coherent partial row can still force confusion). Let $A = \{0, 1, 2, 3\}$ with binary operation

f	0	1	2	3
0	0	2	0	0
1	1	2	1	3
2	0	2	2	0
3	3	0	3	0

and supply

$$\alpha(b, 0) = 0, \quad \alpha(b, 2) = 2, \quad \alpha(b, 3) = 0.$$

Write $r_i = \alpha(b, i)$. The generated domain is $S = \{0, 2, 3\}$, and $h(0) = h(3) = 0$, $h(2) = 2$ is a homomorphism $S \rightarrow A$. Its kernel identifies 0 and 3. The context $f(-, 1)$ adds $2 \equiv 0$ to $\text{Cg}_A(0, 3)$, while $f(1, -)$ adds $1 \equiv 3$. Thus

$$\text{Cg}_A(0, 3) = \nabla_A, \quad \text{Cg}_A(\ker h) \cap S^2 > \ker h.$$

The raw row is not protected. Notice that this universal *row* congruence does not imply universal collapse of the distinguished carrier. The carrier loses exactly the distinction between 0 and 2. Indeed, compatibility and the supplied cells give

$$r_2 = f(r_0, r_1) = f(r_3, r_1) = r_0,$$

so $E \vdash 2 = 0$. The partition

$$\theta = \{0, 2\} \mid \{1\} \mid \{3\}$$

is a congruence, and the constant row with value $[0, 2]$ is an endomorphism of A/θ satisfying the supplied data. This quotient model separates its three classes and proves $\theta_E^A = \theta$.

The missing cell is forced as well. Since $f(1, 0) = 1$ and $f(1, 3) = 3$,

$$r_1 = f(r_1, r_0) = f(r_1, r_3) = r_3 = 0.$$

Therefore all four named row values coincide with the carrier class $[0, 2]$. No external named row value remains in this example.

Examples 6.1 and 6.2 separate two effects of the same kernel. It can spread through A and determine new row values without confusion, or it can feed additional equality back into its original domain and thereby force a carrier quotient. The latter is the local congruence-extension obstruction.

Example 6.3 (External values can coexist with a completion). Let $A = \{0, 1\}$ with $x \vee y = \max(x, y)$ and one entirely unspecified row. The named carrier and the row copy are disjoint in the initial algebra by Theorem 5.3. Both row values are external. Nevertheless, the identity endomorphism of A is a carrier-valued completion. A choice of completion does not turn its chosen values into consequences of the original equations.

6.2 Depth-bounded equality

The preceding results describe full equational consequence. We now resolve that consequence by the depth of terms allowed in a derivation. This small filtration supplies the common example and notation for the companion paper [20]; no general depth theory is needed here.

Constants have depth zero and $\text{dep } f(t_1, \dots, t_k) = 1 + \max_i \text{dep } t_i$. Let $T^{(D)}$ be the set of closed well-sorted terms of depth at most D . On this finite set, $\equiv_D$ is the least equivalence relation containing the ground equations whose sides belong to $T^{(D)}$, and closed under operation applications whenever both resulting terms are in $T^{(D)}$. For the carrier names C_A , put

$$R_A^{(D)} = \equiv_D \upharpoonright_{C_A}.$$

This is a bound on term depth within a derivation, not on its number of steps.

Lemma 6.4 (Ground locality). *Let E be a finite ground equational presentation and $h(E)$ the maximum depth of an axiom side (zero if E is empty). If s, t have depth at most d and $E \vdash s = t$, then $s \equiv_H t$ for $H = \max\{d, h(E)\}$.*

Proof. Let V consist of all subterms of axiom sides and of s, t . Close the given equations under congruence only at operation nodes represented in V . The quotient of V is a partial algebra: if two represented operation nodes have equal argument classes, their result classes agree by closure. For each sort, take these classes as a carrier, adjoining a default element if no class exists. On a tuple represented by a node of V , use its result class; on any other tuple, choose an arbitrary default of the result sort. Interpret constants not in V arbitrarily. Structural induction shows that every term of V evaluates to its class, so this total algebra is a model of E . If the classes of s and t were different, it would separate them, contradicting $E \vdash s = t$. Thus their classes merge. All nodes used by this closure have depth at most H , and each closure step is an allowed bounded-congruence step. This proves the assertion and the usual ground subterm property [18, 19]. $\square$

For each b , let $P_b = \{(a, c) : \alpha(b, a) = c \in \Gamma\}$. Define θ_{val} to be the least native carrier congruence closed under the rule

$$a \equiv_{\theta} a', \quad (a, c), (a', c') \in P_b \quad \implies \quad c \equiv_{\theta} c'. \quad (6.1)$$

It exists, for example by intersecting all congruences satisfying the rule.

Theorem 6.5 (Three structural horizons). *In the typed pure-row fragment,*

$$R_A^{(0)} = \Delta_A, \quad R_A^{(1)} = \theta_{\text{val}}, \quad R_A^{(2)} = \theta_E^A.$$

Horizon one imposes value coherence of supplied cells. Horizon two also imposes homomorphic coherence and kernel feedback.

Proof. There are no equations between depth-zero carrier terms in the presentation, so horizon zero preserves all carrier names. At horizon one, native ground tables and supplied cells are available, but the compatibility equations have sides of depth two. Congruence under native operations imposes native carrier congruence. Congruence under $\alpha(b, -)$ imposes exactly (6.1) on prescribed outputs. No further carrier equality is necessary: interpret depth-one row nodes as separate copies of the quotient by θ_{val} , identifying a row point with a carrier point only where a supplied cell requires it. This finite partial algebra realizes all horizon-one nodes without further carrier identifications. Finally, every axiom side in the pure-row fragment has depth at most two, so Lemma 6.4 gives the horizon-two assertion. $\square$

6.3 Exact quotients of the named interface

Let

$$\mathcal{I} = C_A \cup \{\alpha(b, a) : b \in B, a \in A\}, \quad X_D = \mathcal{I}/(\equiv_D |_{\mathcal{I}}) \quad (D \geq 1).$$

We treat X_D as a quotient of a labelled set. The interface need not be closed under arbitrary native operations. Set

$$\theta_1 = \theta_{\text{val}}, \quad Q_1 = A/\theta_1, \quad \theta_2 = \theta_E^A, \quad Q_2 = A/\theta_2.$$

Let $U_b \subseteq Q_1$ consist of the classes containing at least one supplied input for row b . Rule (6.1) makes the supplied output a well-defined map $v_b : U_b \rightarrow Q_1$; U_b need not be a subalgebra. Over Q_2 , use the stabilized data S_b, h_b, K_b, κ_b already defined.

Theorem 6.6 (Named-interface structure and counts). *The following are exact descriptions as labelled quotient sets.*

- (i) X_1 consists of the central Q_1 and one copy of Q_1 for each row, with the point $x \in U_b$ identified with $v_b(x)$ in the centre. There are no other identifications. Consequently

$$N_1 := |X_1| = |Q_1| + \sum_{b \in B} (|Q_1| - |U_b|). \quad (6.2)$$

- (ii) X_2 consists of the central Q_2 and the row copies Q_2/κ_b , each identified with the centre along $S_b/K_b \cong h_b(S_b)$. Distinct external row parts remain disjoint. Consequently

$$N_2 := |X_2| = |Q_2| + \sum_{b \in B} (|Q_2/\kappa_b| - |S_b/K_b|). \quad (6.3)$$

For an empty row the corresponding contribution is $|Q_2|$.

- (iii) The relation on $\mathcal{I}$ is final at horizon two. Its first stabilization horizon $\delta_E^{\mathcal{I}} \geq 1$ satisfies

$$\delta_E^{\mathcal{I}} = \begin{cases} 1, & N_1 = N_2, \\ 2, & N_1 > N_2. \end{cases}$$

Proof. The horizon-one construction in Theorem 6.5 gives (i). Although different inputs can share a supplied output, each supplied-input class removes precisely one otherwise new row point from the count. For (ii), apply Theorem 5.3. Each row has $|Q_2/\kappa_b|$ image points, of which exactly $|S_b/K_b|$ are central; external points from different rows do not overlap. Ground locality makes this full initial equality already visible at horizon two. Finally, $\equiv_1 |_{\mathcal{I}} \subseteq \equiv_2 |_{\mathcal{I}}$ on the same finite labelled set. These relations are equal if and only if their numbers of classes agree, which proves (iii). $\square$

For a cell, define its first carrier-contact horizon by

$$\chi_E(b, a) = \min\{D \geq 1 : \exists c \in A, \alpha(b, a) \equiv_D c\},$$

with value ∞ when the set is empty. The theorem gives the exact rule

$$\chi_E(b, a) = \begin{cases} 1, & [a]_{\theta_1} \in U_b, \\ 2, & [a]_{\theta_1} \notin U_b \text{ and } [a]_{\theta_2} \in S_b^{\kappa_b}, \\ \infty, & \text{otherwise.} \end{cases}$$

For an empty row, only the last case applies. Similarly, for observed terms s, t , their equality-visibility depth is $\lambda_E(s, t) = \min\{D : s \equiv_D t\}$, or ∞ if they are unequal in the initial algebra.

In Example 6.2, the exact interface partitions are

Horizon	Classes on $\{0, 1, 2, 3, r_0, r_1, r_2, r_3\}$	Count
1	$\{0, r_0, r_3\} \mid \{1\} \mid \{2, r_2\} \mid \{3\} \mid \{r_1\}$	5
2	$\{0, 2, r_0, r_1, r_2, r_3\} \mid \{1\} \mid \{3\}$	3

Thus $\lambda_E(0, 2) = 2$, $\chi_E(b, 1) = 2$, and $\delta_E^I = 2$. The two short derivations in that example use only terms of depth at most two; the horizon-one partition proves that this depth is necessary.

The feedback iteration count is a different quantity. The family in Proposition 5.5 requires $|A| - 1$ strict simultaneous quotient updates, although all its carrier equalities are visible at depth two. A shallow derivation can contain many inference steps, and a structural fixed-point computation can revisit the same shallow constraints.

7 Completion and quotient repair

Initial semantics asks what every model must satisfy. Carrier-valued completion asks for the existence of a much more special model, one in which no named action value leaves the carrier. The distinction is already visible in one protected row.

7.1 Completion is splitting

Let

$$\begin{array}{ccc} S & \hookrightarrow & Q_r \\ h \downarrow & & \downarrow p_r \\ Q_* & \xrightarrow{m} & P \end{array}$$

be a protected row pushout, so m is injective.

Theorem 7.1 (Completion as a split monomorphism). *The partial homomorphism $h : S \rightarrow Q$ extends to an endomorphism $\tau : Q \rightarrow Q$ if and only if the distinguished carrier embedding*

$$m : Q \hookrightarrow P$$

is split: there exists $r : P \rightarrow Q$ with $rm = \text{id}_Q$. In that case

$$\tau = rp_r.$$

Proof. If τ extends h , the maps $\tau : Q_r \rightarrow Q$ and $\text{id}_Q : Q_* \rightarrow Q$ agree on S . The pushout universal property gives a unique $r : P \rightarrow Q$ with

$$rp_r = \tau, \quad rm = \text{id}_Q.$$

Conversely, if r is a retraction of m , then $\tau = rp_r$ satisfies, for $s \in S$,

$$\tau(s) = rp_r(s) = rmh(s) = h(s).$$

□

Theorem 7.2 (Forced closure as image containment). *For a protected row, every row point is forced into the distinguished carrier if and only if*

$$p_r(Q) \subseteq m(Q),$$

equivalently if and only if

$$S^\kappa = Q.$$

When this holds, $m^{-1}p_r : Q \rightarrow Q$ is the unique endomorphism extending h .

Proof. The equivalence with $S^\kappa = Q$ is Theorem 4.1. If $p_r(Q) \subseteq m(Q)$, define $\tau = m^{-1}p_r$. Since both p_r and m are homomorphisms and m is injective, τ is an endomorphism extending h . Every extension identifies K and therefore κ ; if $x \equiv_\kappa s$ with $s \in S$, it must send x to $h(s)$. Since $S^\kappa = Q$, these values determine the whole extension. $\square$

Hence

protection	$\iff$	m monic,	(7.1)
completion	$\iff$	m split monic,	
forced closure	$\iff$	$p_r(Q) \subseteq m(Q)$.	

Forced closure implies completion, and completion implies protection. The converses fail. Example 6.3 gives a completion with external row points. Example 7.6 below gives protection without a completion on the protected carrier.

For several independent rows, strongly amalgamate all protected row factors over the same central Q . A family of simultaneous endomorphism completions exists exactly when the single central embedding into this wide row amalgam splits. Thus the local dictionary extends unchanged to the pure multirow fragment.

7.2 Every repair lies above the forced quotient

Proposition 7.3 (Forced congruence is a universal lower bound). *For every repair $\theta \in \mathcal{R}(E)$,*

$$\theta_E^A \subseteq \theta.$$

Proof. A completion on A/θ is a model of the original ground theory in which the kernel of the named carrier map is exactly θ . Every provable carrier equality must therefore belong to θ . $\square$

Complete tables behave especially well: once the table is well defined on a quotient, protection already gives total endomorphisms, so the least repair is exactly the complete-table congruence of Corollary 5.7. Incomplete tables behave differently.

7.3 No least repair

Example 7.4 (Two incomparable minimal repairs). Let $A = \{0, 1, 2, 3\}$ have one unary operation

$$f(0) = f(1) = f(2) = 0, \quad f(3) = 1,$$

and supply one row

$$\alpha(b, 1) = 2.$$

There is no completion on A . Indeed, if τ were an endomorphism with $\tau(1) = 2$, then $f(1) = 0$ gives

$$\tau(0) = \tau(f(1)) = f(\tau(1)) = f(2) = 0,$$

while $f(3) = 1$ gives

$$2 = \tau(1) = \tau(f(3)) = f(\tau(3)),$$

impossible because $\text{im } f = \{0, 1\}$.

Nevertheless both

$$\theta_a = \{0, 2\} \mid \{1\} \mid \{3\}$$

and

$$\theta_b = \{0\} \mid \{1, 2\} \mid \{3\}$$

are repairs. On A/θ_a the constant map to $[0, 2]$ is an endomorphism satisfying the row; on A/θ_b the identity map satisfies it. The congruences are incomparable and

$$\theta_a \cap \theta_b = \Delta_A,$$

which is not a repair. Each of θ_a, θ_b merges only one pair, so its only strictly finer equivalence relation is the diagonal. They are therefore minimal repairs, and $\mathcal{R}(E)$ has no least element.

This failure is impossible for a complete table and is therefore a specifically incomplete-table phenomenon.

7.4 Repair feasibility is not upward closed

Example 7.5 (A coarser quotient can destroy completion). Let $A = \{0, 1, 2, 3\}$ have the constant unary operation $f(x) = 0$ and supply

$$\alpha(b, 0) = 1, \quad \alpha(b, 3) = 2.$$

The congruence

$$\theta = \{0, 1\} \mid \{2\} \mid \{3\}$$

is a repair. In the quotient, the only endomorphism constraint imposed by the constant operation is that $[0, 1]$ map to itself; one may send $[3]$ to $[2]$.

Now coarsen to

$$\phi = \{0, 1, 3\} \mid \{2\}.$$

The two supplied source inputs 0 and 3 have become equal while their targets 1 and 2 remain distinct. The projected row is not even a function, so ϕ is not a repair. Total collapse is again a repair. Thus $\mathcal{R}(E)$ is not an upper set of $\text{Con}(A)$.

Consequently quotient repair cannot in general be found by a monotone search that simply coarsens until completion appears.

7.5 The repair kernel and the cost of forbidding junk

Define

$$\kappa_E^{\text{NJ}} = \bigcap_{\theta \in \mathcal{R}(E)} \theta. \tag{7.2}$$

By Proposition 7.3,

$$\theta_E^A \subseteq \kappa_E^{\text{NJ}}.$$

The inclusion can be strict.

Example 7.6 (Strict repair-kernel gap). Let $A = \{0, 1, 2, 3\}$ with

$$f(0) = 1, \quad f(1) = 1, \quad f(2) = 1, \quad f(3) = 2,$$

and supply

$$\alpha(b, 0) = 2, \quad \alpha(b, 1) = 1, \quad \alpha(b, 2) = 0.$$

The original carrier is protected. Indeed, adjoin one new element x , set $f(x) = 0$, and define $\alpha(b, 3) = x$. Give $\alpha(b, x)$ any value to make the action total. Then all named-input compatibility equations hold while the four named carrier elements remain distinct. Hence

$$\theta_E^A = \Delta_A.$$

There is no carrier-valued completion on A . For any completion on a quotient, compatibility at input 3 gives

$$[0] = \tau([2]) = \tau(f([3])) = f(\tau([3])).$$

Thus $[0]$ must lie in the image of the quotient operation f , whose named image is represented by $[1]$ or $[2]$. Hence $[0] = [1]$ or $[0] = [2]$. In the first case the supplied rows at inputs 0 and 1 force $[2] = [1] = [0]$ as well. Therefore every repair identifies 0 and 2.

The congruence

$$\theta_* = \{0, 2\} \mid \{1\} \mid \{3\}$$

is itself a repair: the identity endomorphism of A/θ_* satisfies all projected rows. Thus

$$\boxed{\theta_E^A = \Delta_A < \kappa_E^{\text{NJ}} = \theta_*}.$$

The model-theoretic meaning of this gap can be stated without choosing any completion procedure. Add the finite family of disjunctions

$$E^{\text{NJ}} = E \cup \left\{ \bigvee_{c \in A} \alpha(b, a) = c : b \in B, a \in A \right\}. \quad (7.3)$$

This is a first-order theory of total algebras, not an equational theory. Its additional condition concerns only the named action interface; it does not require every element of a model to be named.

Theorem 7.7 (Repairs as kernels of interface no-junk models). *Writing $\ker_A(M)$ for the kernel of the named carrier map into M , one has*

$$\mathcal{R}(E) = \{\ker_A(M) : M \models E^{\text{NJ}}\}. \quad (7.4)$$

Consequently

$$\kappa_E^{\text{NJ}} = \bigcap_{M \models E^{\text{NJ}}} \ker_A(M), \quad \theta_E^A = \bigcap_{M \models E} \ker_A(M) \subseteq \kappa_E^{\text{NJ}}.$$

Proof. If θ is a repair, choose its endomorphism completions and interpret the carrier sort by A/θ and the operator sort by B . This is a model of E^{NJ} with named carrier kernel θ .

Conversely, suppose $M \models E^{\text{NJ}}$ and put $\theta = \ker_A(M)$. For each b and $[a] \in A/\theta$, choose a name c with $\alpha^M(b^M, a^M) = c^M$, and set $\tau_b([a]) = [c]$. The choice of c does not matter because all such names have the same θ -class. Nor does the representative a matter, since equality of its interpretations is preserved by the total operation α^M . Native ground diagrams and compatibility show that τ_b is an endomorphism of A/θ . The supplied cells hold by construction. Thus θ is a repair. The intersection identities follow, the second also being the usual model-theoretic characterization of ground equality. $\square$

A pair in $\kappa_E^{\text{NJ}} \setminus \theta_E^A$ can be kept distinct only by allowing some named action value to be external. This is the precise cost, in carrier distinctions, of requiring interface no-junk. Example 7.6 shows that the cost can be nonzero even when the original carrier is protected.

The no-junk condition is not generally preserved by direct products. Each coordinate can choose a different carrier name for the same missing cell, leaving its product value outside the diagonal named carrier image. Thus a product can intersect carrier kernels while ceasing to satisfy (7.3). This explains why an intersection of repairs need not itself be a repair, without asserting that repairs form a closure system.

8 From independent rows to action laws

The pure-row theorem treats each named operator independently except for the shared carrier quotient. Genuine action laws connect the row copies. The raw-row formula and its feedback operator do not automatically compute the forced quotient of these stronger presentations. In this section θ_E^A always means equality in the full indicated ground theory. We prove the structural consequences directly from that theory.

8.1 Empty data produce a free action forest

Assume $\Gamma = \emptyset$ and no action law beyond carrier compatibility. For a word $w = b_1 \cdots b_n \in B^*$ define

$$W_w(a) = \alpha(b_1, \alpha(b_2, \dots, \alpha(b_n, a) \dots)), \quad W_\varepsilon(a) = a.$$

Theorem 8.1 (Free action forest). *In the empty-row typed theory, the map*

$$A \times B^* \longrightarrow \mathbb{I}_E^A, \quad (a, w) \longmapsto [W_w(a)]$$

is injective. Consequently the initial carrier sort is infinite, and the number of distinct clean chains of length at most d is at least

$$|A| \sum_{k=0}^d |B|^k.$$

Proof. Construct a model with operator sort B and carrier sort $A \times B^*$. Interpret carrier names by (a, ε) . If a native k -ary carrier operation is applied to elements with a common word label w , put

$$f((a_1, w), \dots, (a_k, w)) = (f^A(a_1, \dots, a_k), w),$$

and assign arbitrary values on mixed labels. Define

$$\alpha(b, (a, w)) = (a, bw).$$

The native diagrams hold at the base fiber and carrier compatibility holds because both sides of (2.1) have the same word label b . Evaluation gives

$$W_w(a) \mapsto (a, w),$$

so distinct pairs are separated in a model of E and hence cannot be equal in the initial algebra. $\square$

The theorem identifies an embedded family of clean action chains. It does not assert that these chains exhaust the initial carrier sort: native operations applied to different fibers can produce further terms.

Proposition 8.2 (Finiteness boundary for independent rows). *In the pure typed row fragment, the initial carrier sort is finite if and only if every named ground action term $\alpha(b, a)$ is carrier-valued.*

Proof. If all named action terms reduce to carrier names, structural induction reduces every closed carrier-sort term to a named carrier constant: native operations use D_A , and an action term first reduces its carrier argument to a name and then uses ground-action closure.

Conversely, suppose $x = \alpha(b, a)$ is external. The separation model in Theorem 5.3 realizes x as an element outside the central named carrier. Choose a named b_0 and adjoin infinitely

many fresh elements $x_1, x_2, \dots$. Preserve all native operations on the old carrier and extend them arbitrarily to tuples using new elements. Redefine $\alpha(b_0, x) = x_1$ and put $\alpha(b_0, x_i) = x_{i+1}$; preserve the action on all central inputs. Every pure-row axiom involves the action only on named central inputs (possibly a native term equal to such an input), so these changes preserve all axioms. The terms $x, \alpha(b_0, x), \alpha(b_0, \alpha(b_0, x)), \dots$ have pairwise distinct values in the new model. Hence the initial carrier sort is infinite. $\square$

8.2 Composition folds the forest

Suppose now that B is a finite semigroup with its named multiplication table and add the ground composition scheme

$$\text{Comp} : \quad \alpha(gh, a) = \alpha(g, \alpha(h, a)) \quad (8.1)$$

for named $g, h \in B$ and named $a \in A$.

Proposition 8.3 (Cayley folding). *In the empty-row theory with ground Comp, every nonempty clean chain satisfies*

$$E \vdash W_{b_1 \dots b_n}(a) = \alpha(\overline{b_1 \dots b_n}, a),$$

where the bar denotes the product in B . Moreover, if no other action law is present, then

$$E \vdash W_u(a) = W_v(a') \iff (a = a' \text{ in } A) \text{ and } (\bar{u} = \bar{v} \text{ in } B)$$

for nonempty words u, v .

Proof. Repeated bottom-up use of (8.1) and the multiplication table reduces every clean chain to a single row term. For the converse, use the carrier-sort model consisting of a base copy A_* and copies A_h indexed by $h \in B$, with

$$\alpha(g, a_*) = a_g, \quad \alpha(g, a_h) = a_{gh},$$

and carrier operations acting copywise, with arbitrary values on mixed-copy tuples. Ground Comp and compatibility hold by construction, and different base elements and different product labels remain distinct. $\square$

Thus Comp replaces the free word-indexed forest by a finite Cayley-type diagram of carrier copies

$$A_h \xrightarrow{g} A_{gh}.$$

This is the natural bridge from independent incomplete rows to partial-action and globalization theory.

8.3 Units, inverses, and partial group actions

When B is a group, add the ground schemes

$$\text{Un} : \quad \alpha(e, a) = a,$$

$$\text{Inv} : \quad \alpha(g^{-1}, \alpha(g, a)) = a,$$

in addition to Comp.

For the theory with these additional laws, let $Q = A/\theta_E^A$ and define D_g semantically as the set of $[a] \in Q$ for which $\alpha(g, a)$ equals a carrier name in the full initial algebra. The resulting forced value map is $\tau_g : D_g \rightarrow Q$. These definitions are well defined by congruence; we do not compute them by the pure-row saturation formula.

Proposition 8.4 (Canonical partial group action). *Under ground Un, Comp, and Inv:*

- (i) D_g is closed under all positive-arity native operations. If it is nonempty, it is a subalgebra, and τ_g is a homomorphism on its domain.
- (ii) $D_e = Q$ and $\tau_e = \text{id}_Q$.
- (iii) $\tau_g : D_g \rightarrow D_{g^{-1}}$ is a bijection preserving native operations in both directions, with inverse $\tau_{g^{-1}}$.
- (iv) The exact domain of the composite is

$$\text{dom}(\tau_g \circ \tau_h) = D_h \cap D_{gh},$$

and $\tau_g(\tau_h(x)) = \tau_{gh}(x)$ on this domain.

Thus the forced maps form a partial group action, allowing empty domains. If every $D_g = Q$, they define a homomorphism $B \rightarrow \text{Aut}(Q)$.

Proof. If each $x_i = [a_i]$ belongs to D_g , choose names c_i for their forced images. Ground compatibility gives $\alpha(g, f(\bar{a})) = f(\bar{c})$, which is a named carrier value by D_A . This proves domain closure and preservation of operations. Un gives (ii). If $\tau_g([a]) = [c]$, substitution in ground Inv at a gives $\alpha(g^{-1}, c) = a$. Applying the same argument with g^{-1} proves both inverse identities, establishing (iii).

If $x \in D_h$ and $\tau_h(x) \in D_g$, ground Comp identifies their composite carrier value with $\alpha(gh, x)$, so $x \in D_{gh}$. Conversely, if $x \in D_h \cap D_{gh}$, choose carrier names for $\tau_h(x)$ and $\tau_{gh}(x)$. The same Comp instance states that applying g to the former equals the latter. Hence $\tau_h(x) \in D_g$. This proves the domain identity and the value identity in (iv). Full domains now give automorphisms satisfying the group law. $\square$

Empty domains really occur: take a one-element carrier, $B = C_2$, and no supplied cells. A two-point regular C_2 -orbit, with the named carrier at one point and native operations acting on constant fibers, is a model whose nonidentity action value is external. Hence the corresponding forced domain is empty. Positive arities make its closure condition vacuous; no empty algebra is being required in our category.

This construction connects the forced maps to partial actions and premorphisms [13, 12]. Here the partial maps are consequences of arbitrary ground prescriptions. Their carrier quotient must be taken in the full theory including the group laws.

9 One-sorted couplings and reconstruction

The typed theory isolates the intended action. Some applications deliberately require stronger coupling: carrier and operator names may be identified, or the action symbol may be merged with a native operator operation. These are naturally one-sorted constructions and should be treated separately because their mixed term sector changes global cardinality.

9.1 A free mixed sector survives complete named action data

Consider a one-sorted version with disjoint named copies of A and B , no equation identifying an A -name with a B -name, and a fresh action head α . Supply any complete table on the named block $B \times A$ and any selection of the standard ground laws Un, Ann, Comp, and Inv, whenever the relevant named constants and operations exist.

Proposition 9.1 (Mixed-sector infinitude). *If A and B are nonempty and no cross-copy identification is present, the full one-sorted initial algebra is infinite even though every named*

action value is supplied. In particular, fixing a carrier name a_0 , the following terms are pairwise distinct in it:

$$t_0 = a_0, \quad t_{n+1} = \alpha(a_0, t_n).$$

Proof. It suffices to construct a model separating the tower. Collapse all carrier names to one element c and all operator names to a distinct element o . Interpret the native ground diagrams accordingly. Interpret the complete named action block by $\alpha(o, c) = c$; all standard ground action laws then reduce to equalities involving the operator point o and the carrier point c and can be satisfied by the same choice. Add distinct elements $t_1, t_2, \dots$ and define

$$\alpha(c, c) = t_1, \quad \alpha(c, t_n) = t_{n+1}.$$

Assign all other unconstrained mixed operation values arbitrarily. No standard named action axiom has the carrier element c in the operator position, so the tower is unconstrained. Hence the t_n are separated in a model of the theory. $\square$

This is an encoding artifact rather than an incomplete-action phenomenon: the tower is ill-typed from the intended action perspective. It is the main reason for using the two-sorted core above.

9.2 Coupling can reconstruct multiplication

The one-sorted language is nevertheless useful when cross-identification is intentional. Let the carrier be the additive group $A = \mathbb{Z}/n\mathbb{Z}$ and let $B = M_n$ be the multiplicative monoid on the same underlying set, with distinct names initially. Add equations identifying like-named residues; call this *by-name coupling*.

Proposition 9.2 (Two reconstruction regimes). *Consider only the native diagrams, by-name coupling, and the additional laws listed in the relevant clause below; no extra conflicting cells are imposed. Then:*

- (i) *if α is identified as a signature symbol with the multiplication of M_n , the native ground tables alone reduce every closed term to a named residue;*
- (ii) *if α remains separate but the ground unit law*

$$\alpha(1, a) = a$$

and scalar-additivity law

$$\alpha(b_1 + b_2, a) = \alpha(b_1, a) + \alpha(b_2, a)$$

are present on named residues, then

$$E \vdash \alpha(b, a) = ba$$

for every named b, a , and every closed term again reduces to a named residue.

In either regime the named expansion is the canonical $\mathbb{Z}/n\mathbb{Z}$ structure.

Proof. In the merged regime, by-name coupling turns every action on named constants into a row of the complete multiplication table; structural induction then reduces all terms by the native tables.

In the separate regime, repeated scalar additivity and the unit row give

$$\alpha(k, a) = ka$$

for residues represented by sums of k copies of 1. The final cyclic recurrence yields the zero row. Hence every named action value is canonical, after which structural induction again reduces all closed terms. The standard ring expansion is a separating model for distinct residue names. $\square$

This example is intentionally outside the pure typed theorem: the added name identifications make operations from the two original signatures interact on one common carrier.

10 Finite computation and reproducibility

The carrier and named-action diagnostics are finite even when the initial algebra is infinite. For a finite set of observed terms O , take

$$V = \text{Sub}(E) \cup \text{Sub}(O),$$

where $\text{Sub}(E)$ means the subterms of both sides of every axiom. Congruence closure on the represented operation nodes computes exactly the equality relation on O , by Lemma 6.4. There is no need to enumerate the entire term universe $T^{(2)}$.

The same finite construction can be filtered by depth. At horizon D , retain nodes of depth at most D and activate only equations whose two sides are retained. Comparing the successive restrictions to $\mathcal{I}$ gives λ_E , χ_E , and $\delta_E^{\mathcal{I}}$ where defined. Recording the premises of each merge can produce equality derivations. The finite partial quotient in the locality proof also gives a total separating model for any unmerged query. These are standard uses of ground congruence closure, not a new general equality decision procedure.

The structural feedback operator offers another route to the carrier quotient. A raw row pushout can be queried on its finite carrier and row tables without constructing its possibly infinite mixed-term part. Iterate the resulting carrier kernels as in Proposition 5.4. After stabilization, generate each S_b , compute h_b and $\text{Cg}_{Q_E}(\ker h_b)$, and read off the forced domains and interface counts. The agreement between this route and direct ground congruence closure is an explicit consequence of Theorems 5.2 and 5.3.

Completion requires a separate finite search. For a candidate quotient $Q = A/\theta$, enumerate maps $Q \rightarrow Q$, retain native endomorphisms, and check the supplied values. Rows can be treated independently in the pure fragment. With composition or group laws, impose the corresponding constraints jointly on the chosen maps. This elementary search is useful for small counterexamples; no polynomial bound for general repair search is asserted here.

The supplementary source archive contains a self-contained Python script, `verification/verify_article.py`, and its JSON results. It checks the displayed operation tables, congruences, row kernels, interface partitions, repair witnesses, and the sharp feedback family on finite instances. It also compares two implementations of the named-interface relation on a bounded collection of small presentations: input-subterm congruence closure and row-copy propagation with carrier feedback. These computations are reproducible evidence for the examples; the universal statements rest on the proofs above.

This division of tasks sets the scope of the companion programme. Paper II [20] studies presentation-sensitive visibility beyond the small filtration used here. The planned Paper III will relate computational representations to their equality semantics and specify independently checkable certificates. The associated software project is QKF-Certifier. The present paper does not identify that implementation with a complete solver for every completion and repair problem considered here.

11 Relation to previous work

Initial algebras and algebraic specification. Initial-algebra semantics is classical [1, 2]. The distinction between new values and new equalities among old values is standard in sufficient completeness, hierarchical consistency, and protecting extensions [5, 3, 4]. Our ground-action closure is a restricted no-junk question, while θ_E^A measures exact confusion on the distinguished carrier. The contribution is the exact interaction between these effects for arbitrary incomplete action rows.

Partial algebras. Encoding finite data by partial algebras, quotienting by generated congruences, and freely completing partial structures are classical constructions [6, 7, 8]. They provide natural infrastructure for the finite row-copy picture. We do not claim a new general completion theory of partial algebras.

Congruence extension and amalgamation. The condition

$$\text{Cg}_Q(K) \cap S^2 = K$$

is a particular congruence-extension condition. Its relation to stability of embeddings under pushout belongs to the classical amalgamation/codescent framework; compare [10]. Minimal congruence extension conditions also occur in double-pushout treatments of partial and total algebras [11]. What is specific here is the identification of K as the kernel created by an incomplete action row and the use of its ambient saturation to compute both forced carrier equality and forced missing action values.

Partial actions and globalization. The closest neighboring work is the globalization theory of already-defined partial actions. Khrypchenko and Novikov associate generalized amalgams of copies of an algebra to a semantic partial group action and characterize globalizability through embeddability [12]. Partial monoid and semigroup actions admit related globalization constructions [13, 14, 15]. Our input occurs one logical step earlier: arbitrary ground cells need not define a partial map or a partial homomorphism and may themselves force a quotient of the predefined carrier. For independent rows the quotient is characterized by Theorem 5.2. With group laws, the canonical partial action of Proposition 8.4 instead uses the carrier quotient of the full augmented ground theory. The two settings must not be interchanged.

Incompletely specified operations. The distinction between a partial operation and a total operation whose output is merely unspecified has been studied explicitly in finite clone theory [16, 17]. The present paper uses the same semantic distinction but asks a different question: how incomplete ground values interact with native algebraic structure under initial equational semantics.

Ground congruence closure. The bounded proof-horizon discussion and the finite diagnostic algorithm use standard ground congruence closure and subterm locality [18, 19]. The exact two-stage interface description is specific to the present shallow ground signature. The broader presentation-sensitive question is treated in the companion manuscript [20]. In particular, the number of quotient-feedback rounds is not identified with proof depth or with implementation running time.

12 Conclusion

The pure-row presentation has an exact semantic decomposition. First take the least carrier congruence for which every supplied row becomes homomorphically coherent and kernel-stable. Then the ambient extension of each row kernel determines precisely which remaining inputs acquire carrier values. The resulting quotient and domains are consequences of the equations, without any choice of a completion.

The local-to-global argument also determines the entire equality relation on the named interface. Its horizon-one quotient records value coherence; its horizon-two quotient records the full row geometry. Their class counts give an exact stabilization test, while the sharp feedback family shows why this proof-depth bound does not bound the number of structural updates.

Completion asks for an additional property: a retraction of the row pushout onto the distinguished carrier. Its quotient solutions need not have a least element and need not survive arbitrary coarsening. Their intersection has a separate meaning as the carrier equality forced by models with no external named action values. The strict repair-kernel gap makes the difference from unrestricted initial semantics explicit.

Additional composition and group laws connect rows and require their own full-theory quotient. One-sorted coupling can introduce mixed terms or reconstruct operations, depending on the equations imposed. These boundaries specify where the pure-row characterization applies and which new interactions a more general algorithm must account for. They also provide a common foundation for the companion studies of equality visibility and certificate-producing computation.

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